



**Technological Institute of the Philippines  
College of Computer Studies**

## **FairShare A**

AN ACTIVITY SUBMITTED IN PARTIAL FULFILLMENT  
OF THE REQUIREMENT OF MOBILE COMPUTING

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February 2026

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## Problem Statement

People who are budget-conscious track their financial activity through financial apps. These apps let you log your monthly income and expenses. However, when people go on group events between friends (like road trips, vacations, out-of-country trips, etc.), some expenses that happen during these events are forgotten to be logged and leave the people with a feeling of money lost when, in reality, they have their money in the form of IOUs. These "context gaps," where users' ledgers are not updated on social spending, leaving users wondering where their money went.

Another problem that we see is the "awkwardness factor," where after the event, people forget that they owe someone some money, and the one who lends the money feels like they cannot get their money back from those that they lent some money to.

And the last problem that we see is the problem of expense management for a group of people. Where the problem lies is in the manual calculation and tracking of expenses, especially if the expense arrangement is one person pays for one thing and another for another thing, and when someone ends up having to pay out of pocket for a sudden expense.

## Objectives

To develop **FairShare**, a multi-pillar financial management application that bridges the gap between personal budgeting and social expense tracking, ensuring financial transparency and reducing social friction within groups.

### Specific Objectives

1. **To Integrate Personal and Social Ledgers:** Create a unified system where users can log personal daily expenses and seamlessly "pivot" shared costs into group sessions, eliminating the "context gap" in their financial records.
2. **To Implement a Real-Time Collaborative "Lobby":** Utilize Firebase Realtime Database to provide a synchronized environment where group members can instantly view, add, and verify shared expenses during events.
3. **To Automate Complex Debt Simplification:** Develop a minimum debt path algorithm that calculates the most efficient way to settle group balances, reducing a "web of debt" into the fewest possible transactions.
4. **To Mitigate Social Friction via Automated Notifications:** Design a "Nudge" system using Firebase Cloud Messaging (FCM) that uses pre-written, polite, or humorous templates to remind peers of outstanding IOUs without the need for manual, awkward confrontation.
5. **To Provide Persistent Peer-to-Peer Tracking:** Establish a dedicated 1-on-1 Mode that maintains a rolling balance for long-term "friendly debts," ensuring that small, recurring loans between individuals are never forgotten.

# Target Users

## 1. The Social Spender (College Students / Barkada)

This user is the heart of the "Social Friction" problem. They deal with high-frequency, low-value transactions where "asking for money" feels petty but necessary.

- **The Vibe:** Informal, mobile-first, and budget-conscious.
- **FairShare Interaction:**
  - **Group Mode:** Used for weekend "Inuman," road trips, or group study snacks.
  - **IOU Mode:** Tracking ₱50 coffee or ₱100 fare loans.
- **Refined Needs:**
  - **Hyper-Local Integration:** Deep-linking to **GCash** and **Maya**. When they tap "Settle," it should ideally open the e-wallet app with the amount pre-filled.
  - **Gamified Nudges:** Instead of "Pay me back," the nudge could be, "Round 2 is on you if you don't settle this ₱50!"
  - **Social Visibility:** A "Hall of Fame" (or Shame) showing who is the fastest to settle their debts.

## 2. The Reimbursable Pro (Working Professionals / Freelancers)

This user needs to prove where their money went. They aren't just splitting a bill; they are managing a "work vs. life" balance.

- **The Vibe:** Organized, efficiency-driven, and hates manual data entry.
- **FairShare Interaction:**
  - **Personal Tracker:** Essential for separating personal spending from project-related expenses.
  - **Group Mode:** Used for "Team Lunch" or "Office Pitch" contributions.
- **Refined Needs:**
  - **OCR Receipt Scanner:** The ability to snap a photo of a BIR-official receipt and have the app auto-extract the total and tax for the Personal Tracker.
  - **One-Tap Export:** A "Generate Report" button that creates a PDF/CSV for the HR department or their personal accountant.
  - **Corporate Silos:** Multiple ledgers so their "Client A" expenses don't mix with their "Friday Night Drinks" expenses.

### 3. The Digital Nomad (Expats / International Travelers)

This user deals with "mathematical friction" on a global scale. They are often confused by exchange rates and conversion fees.

- **The Vibe:** Global, tech-savvy, and precision-oriented.
- **FairShare Interaction:**
  - **IOU Mode:** Tracking debts across borders (e.g., "I paid in USD, you pay me back in PHP").
- **Refined Needs:**
  - **Live FX API:** Integrating a real-time exchange rate (e.g., via ExchangeRate-API) so the 1-on-1 mode updates the "Net Owe" as currency values shift.
  - **Smart Rounding:** In countries like the Philippines, where small centavos are rarely used in cash, the app should offer a "Round to nearest ₱1 or ₱5" toggle.
  - **Cross-Border Remittance Links:** Providing info/links for Wise or Binance P2P for international settlement.

## Features

1. **User Account Management**
  - Makes account creation possible and includes the social aspect of the application.
2. **Personal Finance Tracker**
  - Focuses on logging and managing daily expenses for the self and allows for preset and customizable tags to create custom labels on items spent.
3. **Group Mode Finance Tracker**
  - Designed for payments shared among groups to quickly visualize the amount to pay each user for the occasion, minimizing the issues that come with following up loan payments.
4. **IOU Tracker and Pinger**
  - Manages or creates alerts for payment tracking between users.

## Scope and Limitations

The application is designed for users ranging from adolescents to late adults. It intends to primarily cater to individuals concerned with general expense management, engage in peer-to-peer lending, and participate in group-based expenses. For specific features, a network connectivity is required for the application to be functional—otherwise

It is necessary to establish that the mobile application is intended to only be developed for Android mobile devices—other mobile OS and web-based applications adaptations of the system are currently not in the scope of the project's development timeline. FairShare is not and will not be affiliated with any financial institution—it does not facilitate real-time financial transactions with other services and applications, focusing solely on the features.